EXPERIMENT 8

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Experiment 6 – Design of a Parameterized PL/SQL Stored Procedure for Employee Count by Gender (CData, Toddle)

Aim

To design and implement a stored procedure that accepts gender as an input parameter and returns the total number of employees based on the gender passed, thereby demonstrating procedural programming and parameterized logic in PL/SQL.

Objective

- To understand the structure and components of a stored procedure.
- To implement IN and OUT parameters in PL/SQL procedures.
- To pass dynamic input values to stored procedures.
- To apply aggregate functions for data analysis.

Software Requirements

Database Management System:
- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

Database Administration Tool / Client Tool:

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• Oracle SQL Developer (for Oracle XE)
• pgAdmin (for PostgreSQL)

Problem Statement

Organizations often require flexible reporting mechanisms to analyze workforce distribution based on gender for compliance and internal analytics.

Practical / Experiment Steps

• Accepts gender as an input parameter
• Counts the total number of employees matching the given gender
• Returns the count as output
• Create and execute the procedure using dynamic input values

Procedure

1. Open Oracle SQL Developer or pgAdmin and connect to the database.
2. Create an employee table with required fields (emp_id, emp_name, gender, salary).
3. Insert sample records into the table.
4. Create a parameterized stored procedure that accepts gender as input.
5. Use COUNT() function inside the procedure to calculate employee count.
6. Execute the procedure by passing different gender values.
7. Display the output using appropriate statements.

Input / Output Details

Input:
Employee table:
• emp_id (Integer)
• emp_name (Varchar)

• gender (Varchar)
• salary (Numeric)

Parameter passed:
• Gender (Male/Female)

Step-wise Output

Step 1 – Create Employee Table
Step 2 – Insert Data into Employee Table
Step 3 – Create Stored Procedure
Step 4 – Execute Procedure with Input Parameter
Step 5 – Display Output (Employee Count)

Learning Outcome

• Understand the structure and components of a stored procedure.
• Implement IN and OUT parameters in PL/SQL procedures.
• Pass dynamic input values to stored procedures.
• Apply aggregate functions within procedures for data analysis.
• Develop reusable database logic for real-world business scenarios used in companies like CData and Toddle.

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Creating the table

```
CREATE TABLE employees (
    emp_id SERIAL PRIMARY KEY,
    emp_name VARCHAR(50),
    gender VARCHAR(10),
    salary NUMERIC(10,2)
);

INSERT INTO employees (emp_name, gender, salary) VALUES
('Amit', 'Male', 30000),
('Riya', 'Female', 35000),
('John', 'Male', 28000),
('Sneha', 'Female', 40000);

-- Check data
SELECT * FROM employees;
```

Procedure

```
-- ✅ Procedure
CREATE OR REPLACE PROCEDURE get_employee_count_by_gender(
    IN p_gen VARCHAR(20),
    OUT count_emp INT,
    INOUT status VARCHAR
)
LANGUAGE plpgsql
AS $$
BEGIN
    SELECT COUNT(*) INTO count_emp
    FROM employees
    WHERE gender = p_gen;

    status := 'SUCCESS';
END;
$$;    -- 🔴 IMPORTANT: semicolon here
```

Calling

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```
DO $$
DECLARE
    gen VARCHAR(20) := 'Male';
    count_by_gen INT;
    status VARCHAR(20) := 'Fail';
BEGIN
    CALL get_employee_count_by_gender(gen, count_by_gen, status);

    RAISE NOTICE 'GENDER IS % YOUR COUNT IS % AND STATUS IS %',
        gen, count_by_gen, status;
END;
$$;
```

Output

Output:

```
CREATE TABLE
INSERT 0 4
 emp_id | emp_name | gender |  salary
--------+----------+--------+----------
      1 | Amit     | Male   | 30000.00
      2 | Riya     | Female | 35000.00
      3 | John     | Male   | 28000.00
      4 | Sneha    | Female | 40000.00
(4 rows)

CREATE PROCEDURE
DO

psql:commands.sql:46: NOTICE:  GENDER IS Male YOUR COUNT IS 2 AND STATUS IS SUCCESS
```